

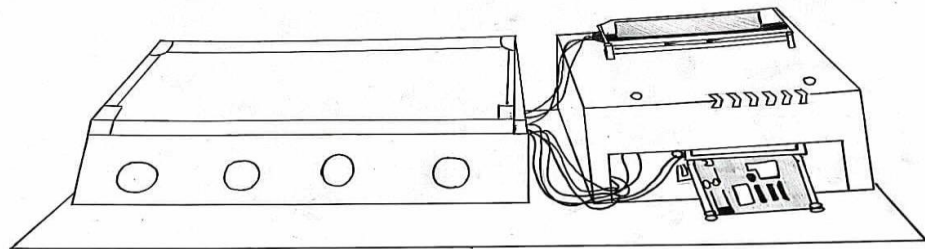
Form 1 Innovative Project for Society (IPS)

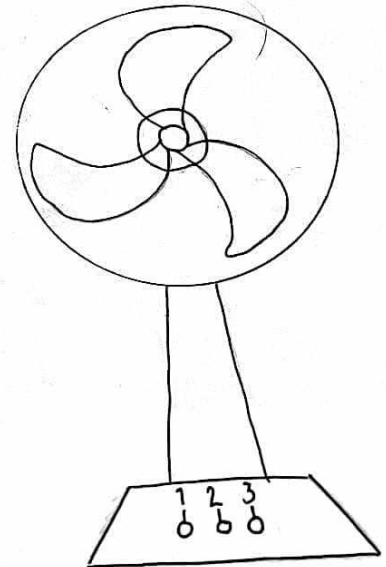
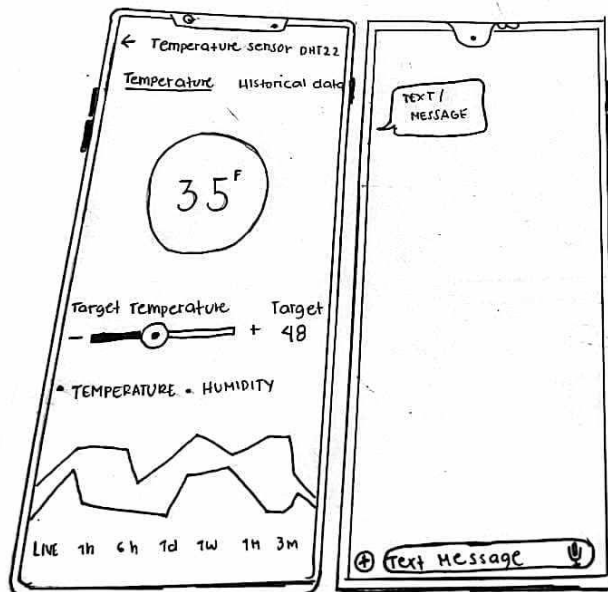
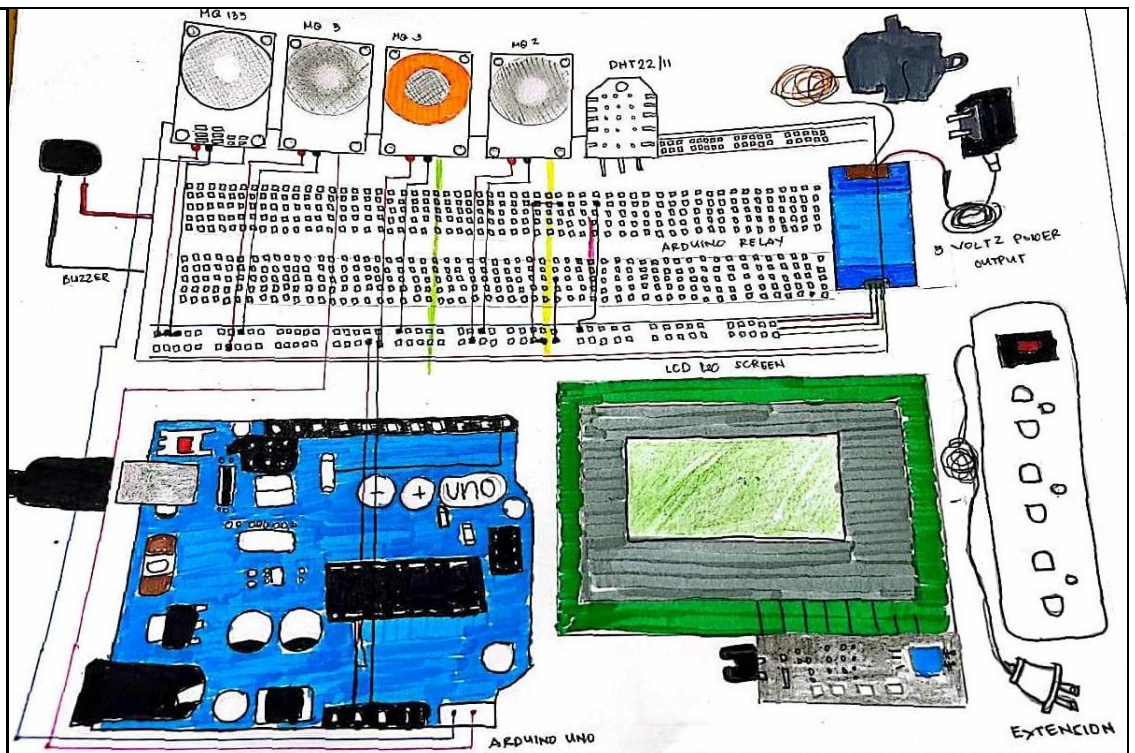
Title: “Triosphere Guardian: Revolutionizing Air Quality Monitoring with Advanced IoT Technology”

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Diagram:





Mechanism: (How it works)	The "TrioSphere Guardian" device operates through a combination of advanced technologies to ensure efficient monitoring of atmospheric gas concentrations. Here is how the device works based on its key features: • Remote Access: The device is designed to allow remote access, enabling users to monitor gas levels from anywhere at any time. This feature ensures that users can stay informed about the air quality in specific areas without the need for physical presence. • Integration with Disaster Risk Management Agencies: The TrioSphere Guardian device is integrated with disaster risk management agencies to facilitate quick response and coordination in case of emergencies. By sharing real-time data with these agencies, the device contributes to better disaster preparedness and response strategies. • Remote Access: Battery Operated and Solar Powered: The device is designed to be energy-efficient by being battery operated and solar powered. This ensures continuous operation even in areas with limited access to electricity. • Remote Access: The device has data logging capabilities that record and store historical gas concentration data. This feature allows users to analyze trends over time, identify patterns, and make informed decisions based on past data.
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Purpose:	The study titled “Triosphere Guardian: Revolutionizing Air Quality Monitoring with Advanced IoT Technology” aims to determine the proportion of dangerous gases like ammonia, methane, and carbon monoxide in the air at a given place by installing an alarm and monitoring system. This device will aid in the prompt response to possible calamities by enabling real-time monitoring of dangerous gases. The researchers will also include an SMS system to spread information when it detects high concentrations of dangerous gases in the atmosphere. This feature will be useful for informing authorities and/or individuals about hazardous gas levels in real-time, enabling them to take appropriate precautions immediately. Moreover, this innovative device incorporates battery-operated and solar-powered features into its monitoring system. The purpose of using these technologies is to ensure continuous and reliable operation of the air quality monitoring device, even in areas where access to electricity may be limited or unreliable. Thus, it has a data logging feature that enables users to track and analyze the records of hazardous gases detected in a specific area.
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Proposed Budget:	Materials	Quantity	Estimated Price
	Node MCU ESP8266	1	₱1680
	Channel Relay Module	1	₱42
	MQ-4 Methane Sensor	1	₱99
	MQ7- Carbon Monoxide Sensor	1	₱105
	MQ-2 Ammonia Sensor	1	₱99
	Arduino Nano	1	₱1337
	Arduino GSM Shield V1	1	₱1600
	PCB Board	1	₱25
	Servo Motor	1	₱89
	12-V Power Supply	1	₱115
	Resistor	10	₱19
	Wire	20	₱200
	Screw	1	₱30
	Solar	1	₱100
	Battery	1	₱500